MILITARY SPECIFICATION SHEET

Connector, Electrical, Bayonet Coupling, Environmental Resisting, Umbilical, General Specification for

This amendment forms a part of Military Specification Sheet MIL-C-81582B, dated 5 November 1975, and is approved for use by all Departments and Agencies of the Department of Defense.

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2.1 In list of Federal Specifications;

Delete: QQ-N-290 and QQ-S-764 and the accompanying titles.

Add: TT-S-735 Standard Test Fluids, Hydrocarbon

In the list of Military Specifications;

Delete: MIL-W-5086/1 and the accompanying title

Add: MIL-A-8243 Anti-icing and Deicing and Defrosting Fluid

MIL-W-22759/12 Wire, Electric, Fluoropolymer - Insulated, Extruded TFE, Nickle-Coated Copper, 600 Volt

MIL-C-25769 Cleaning Compound, Aircraft Surface, Alkaline Waterbase

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2.2 Add: American Society for Testing and Materials.

ASTM A484 Specifications for General Requirements for Stainless and Heat-Resisting Wrought Steel Products (Except Wire).

ASTM A582 Specification for Free Machining, Stainless Steel Bars, Hot-Rolled or Cold-Finished

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

3.3.3.1 Delete and Add:

Contact plating shall be in accordance with MIL-C-39029.

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3.3.5.3 Revise to read:

Unless otherwise specified (see 3.1) aluminum parts shall be cadmium plated to withstand 500 hour salt spray (see 4.6.19). A preliminary plating of another metal is permissable. The resultant finish shall be olive drab (light to dark) and shall be electrically conductive.

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Table 1A Last column;

Delete: MIL-W-5086/1-12

Add: MIL-W-22759/12-10

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Add new paragraph:

3.6.10.1 Retention system fluid immersion. When tested as specified in 4.6.11.1, the insert assemblies shall meet the requirements of contact retention (see 3.6.6). Effect of the fluids on resilient sealing members shall not be a consideration of this test.

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Add to Table XVII:

Dielectric After Fluid Immersion	Requirement Paragraph	Test Paragraph	Test Sample	
			1	2
Examination of product	3.1, 3.7, 3.8	4.6.1	X	X
Dielectric withstanding voltage at sea level	3.6.8	4.6.9.1	X	X
Coupling torque	3.6.5	4.6.6	X	X
Fluid immersion	3.6.10	4.6.11	X	X
Dielectric withstanding voltage at sea level	3.6.8	4.6.9.1	X	X
Coupling torque	3.6.5	4.6.6	X	X
Examination of product	3.1, 3.7, 3.8	4.6.1	X	X

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Table XVII (Continued)

Retention System After Fluid Immersion	Requirement Paragraph	Test Paragraph	Test Sample	
			1	2
Examination of product	3.1, 3.7, 3.8	4.6.1	X	X
Retention system fluid immersion	3.6.10,1	4.6.11	X	X
Contact retention	3.6.6	4.6.7.2	X	X
Examination of product	3.1, 3.7, 3.8	4.6.1	X	X

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4.6.9.2 First sentence, delete "Method 105, Test Condition C, MIL-STD-202; replace with "Method 3001, Test Condition IV, MIL-STD-1344.

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4.6.11 Delete, in it's entirety, present paragraph and items (a), (b) and (c).

Add new paragraphs:

4.6.11 Fluid immersion. Connector samples shall be subjected to the test procedures and fluids as specified in Method 1016 of MIL-STD-1344 (one sample per fluid). Following the fluid immersion cycles the connectors shall be tested for coupling torque as specified in 4.6.6, and dielectric withstanding voltage at sea level as specified in 4.6.9.1 within 3 hours.

4.6.11.1 Retention system fluid immersion (see 3.6.10.1). Unmated connectors with contacts removed shall be immersed in the fluids specified in Method 1016 of MIL-STD-1344 (one sample per fluid) for two hours at room temperature. After removal, excess fluid shall drain from the connectors for 4 hours and the contacts reinstalled. Following the test, the connectors shall be subjected to contact retention as specified in 4.6.7.

4.6.12 Add, after last sentence, (see 4.6.19)

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4.6.19 Delete present paragraph, add:

4.6.19 Salt spray (dynamic test). The wired, assembled plugs and receptacles shall be wired and unmated 50 cycles at the rate of 300 cycles per hour maximum. The mating and unmating shall be accomplished so the plug and receptacle are completely separated during each cycle. The connectors shall then be subjected to the salt spray test in accordance with Method 1001 of MIL-STD-1344. The following details and exceptions apply:

a. The connectors shall be tested for 452 hours mated followed by 48 hours unmated.

b. The connectors shall not be mounted but shall be suspended from the top of the chamber using waxed twine or string, glass rods or glass cord.

c. Wire ends must be protected to prevent salt migration.

After salt spray exposure the remaining number of durability cycles specified in 4.6.12 shall be completed.

6.4.1 Application for qualification. Revise to read:

The activity responsible for the QPL for MIL-C-81582 is the Naval Air Systems Command. The Naval Avionics Center has been designated by the Naval Air Systems Command as agent for establishment of QPL. Requests for information pertaining to and applications for qualification should be addressed to:

Commanding Officer
Naval Avionics Center
6000 E. 21st Street
Indianapolis, IN 46218

6.5 Table XX is now Table XXI.

Custodians:
Army - CR
Navy - AS
Air Force - 99

Preparing Activity:
Navy - AS
Project No. 5935-3212

Review Activities:
Army - AR
Navy - EC
Air Force - 11, 15, 99
DSA - ES

User Activity:
Navy - SH

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